

TABLE I
Job Responsibilities, Skills and Qualifications of a Research Engineer

Typical Job Responsibilities	Typical Skills Expected	Qualifications Expected
1. Develop concept in 3D and conduct FEA for optimising the design 2. Selection of sensors for various applications 3. Mechanical-electronic interface knowledge 4. Good knowledge in servo/stepper selection and motion controllers 5. Component and system-level knowledge in reliability testing 6. Ability to develop modules or technology concepts that involve electro-mechanical systems, interfacing with other signal-processing modules 7. Programming knowledge 8. Systems engineering with ability to understand and design performance characteristics of sub-systems/sub-assemblies, their interdependence and their influence on overall system performance	1. Strong in solid mechanics, mechatronics 2. Knowledge in 3D, preferably SolidWorks, Cosmos/Ansys, etc 3. Good knowledge in sensors and their applications 4. Ability to work in global environment 5. Self starter with good interpersonal and networking skills 6. Focus on quality and result orientation 7. Optimistic, can-do attitude 8. Excellent verbal and written communication skills	1. Masters in mechanical/mechatronics with 0-2 years of experience in mechatronics systems development 2. Excellent technical knowledge and competence to handle the responsibilities listed 3. Ability to learn and keep abreast of related emerging technologies 4. Strong execution skills and innovation mindset

TABLE II
Job Responsibilities, Skills and Qualifications of an Assistant Maintenance/Service Engineer

Typical Job Responsibilities	Typical Skills Expected	Qualifications Expected
1. Perform preventive maintenance of filling and packaging lines 2. Conduct customer training 3. Troubleshooting and customer support 4. Installation and commissioning of filling machines/distribution systems 5. Travel in the region and abroad (for training and customer support)	1. Experience and knowledge of packaging and wrapping technology would be an advantage 2. Knowledge or experience of food processing technology/industry would be an advantage 3. Knowledge of PLC programming and electrical troubleshooting would be an advantage 4. Analytical and problem-solving skills 5. Result oriented 6. Team player 7. A detail-oriented and methodical individual 8. Ability to work under pressure for longer duration 9. Willingness for extensive travel (20-25 days per month)	1. Diploma/degree in mechatronics (preferably) electronics/electrical. 2. 2-4 years of industrial experience in capital equipment maintenance post degree/diploma

TABLE III
Job Responsibilities, Skills and Qualifications of a Design Engineer

Typical Job Responsibilities	Typical Skills Expected	Qualifications Expected
1. Responsibilities include development of algorithms and real-time software in close consideration of object dynamics 2. Management of software releases 3. Troubleshooting system issues related to motion control 4. Work in close cooperation with mechanical group to participate in process of component selection 5. Writing specifications 6. Assisting in writing documentation	1. Familiarity with semiconductor industry 2. Experience in development of motion control application using proprietary development environments like MEI and Delta Tau 3. Knowledge in area of real-time systems, mechatronics, robotics, symbolic computation and motion controllers 4. MATLAB/Simulink Communication: 1. Excellent written, verbal and presentation skills 2. Ability to clearly communicate his/her idea, write white papers and present to large audience 3. A self-motivated, independent contributor who can work with minimum supervision 4. Previous experience in development of complex system in team environment	1. BE/B.Tech in any discipline or mechatronics and ME in mechatronics is must

pace and every innovation involves advancements in either software or hardware. There is tremendous scope for automation in industries, which requires research in these combined sciences. There are terrific career opportunities for students who would like to pursue a career in this direction. We cannot predict how advanced the technology will be five years

from now, but we can confidently predict that the team that designs it will include computer programmers and electronic embedded systems experts."

"There is a lot of scope in the mechatronics sector," believes Dr S.K. Saha, Naren Gupta chair professor, Department of Mechanical Engineering, IIT Delhi. He says, "Today almost

everything we use in our daily life, like mobile phone (it has a motor for vibration and of course other electronic items inside to show the display, communicate with other users, etc), washing machine, motor car, etc, is a combination of mechanical and electrical/electronic items. Hence, the knowledge in that area is a must in case somebody wants to work in companies which